The Power of Vaccination

From Smallpox Eradication to COVID-19 in Record Time

Scope: 1796–2020

Audience: Public health professionals & policy analysts

Source: Our World in Data (ourworldindata.org/vaccination)

From 100 Years to Under 1 Year: Vaccine Development Is Accelerating

Time from pathogen identification to vaccine licensure

Typhoid fever — years (1884→1989)

Meningitis — 92 years (1889→1981)

Polio — 47 years (1908→1955)

Measles — 10 years (1953→1963)

Ebola — 43 years (1976→2019)

COVID-19 — **<1 year (2020→2020)**

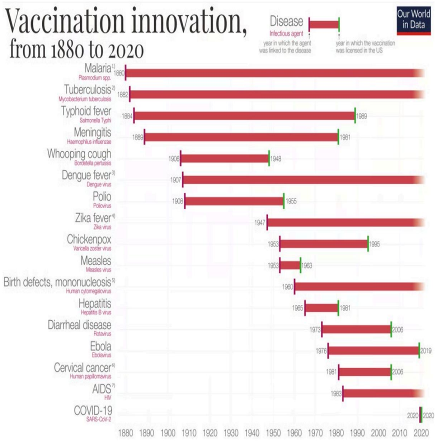


Figure reference: Vaccination innovation timeline (1880–2020)

105 years

Earliest example in brief

Genome sequencing • mRNA • VLPs

Source: Our World in Data (ourworldindata.org/vaccination)

100% Case Reduction Achieved for Diphtheria, Polio, and Smallpox in the US

Disease	Pre-vaccine cases (per million/yr)	Post-vaccine (per million/yr)	Case reduction	Death reduction
Diphtheria	158	0	**100%**	**100%**
Smallpox	250	0	**100%**	**100%**
Acute polio	141	0	**100%**	**100%**
Measles	3,044	0.2	99.99%	100%
Pneumococcal	233	139	40.5%	31.3%
Hepatitis A	465	51	89%	88.7%
Acute hepatitis B	273	44	83.9%	83.6%

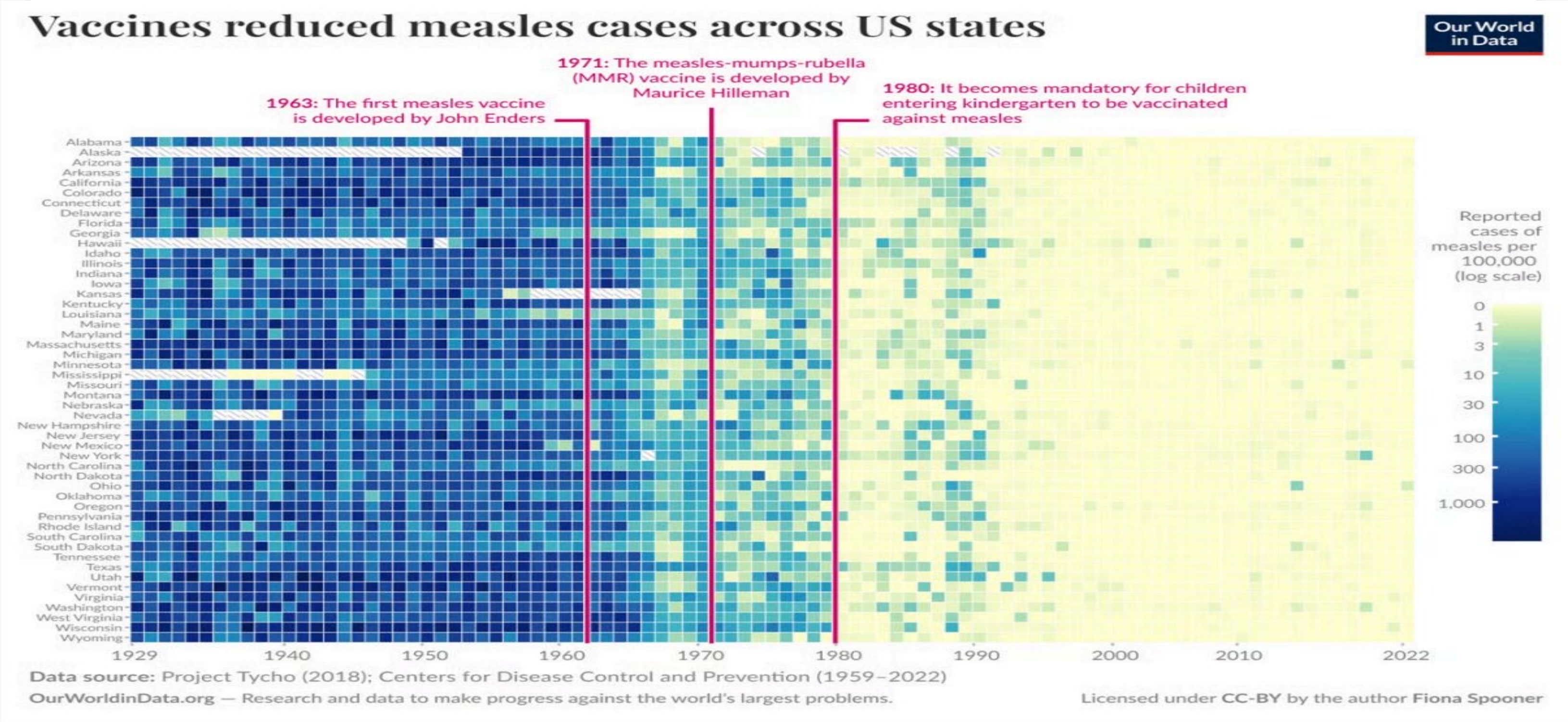
Diphtheria • Smallpox • Acute polio

Pneumococcal: 40.5% cases • 31.3% deaths

Source: Our World in Data (ourworldindata.org/vaccination)

Measles Cases Collapsed Across All 50 US States After Vaccine Mandates

Heatmap-style visualization: measles cases per 100,000 by state (1929–2022).



1963 First measles vaccine (John Enders)

Source: Our World in Data (ourworldindata.org/vaccination)

1971 MMR vaccine developed (Maurice Hilleman)

1980 Kindergarten vaccination mandatory

Smallpox: The Only Human Disease Eradicated Through Vaccination

Milestones (from brief)

1959 WHO resolution to eradicate smallpox

1967 Intensified global eradication campaign launched

1977 Last natural case (Somalia)

1980 Eradication certified worldwide

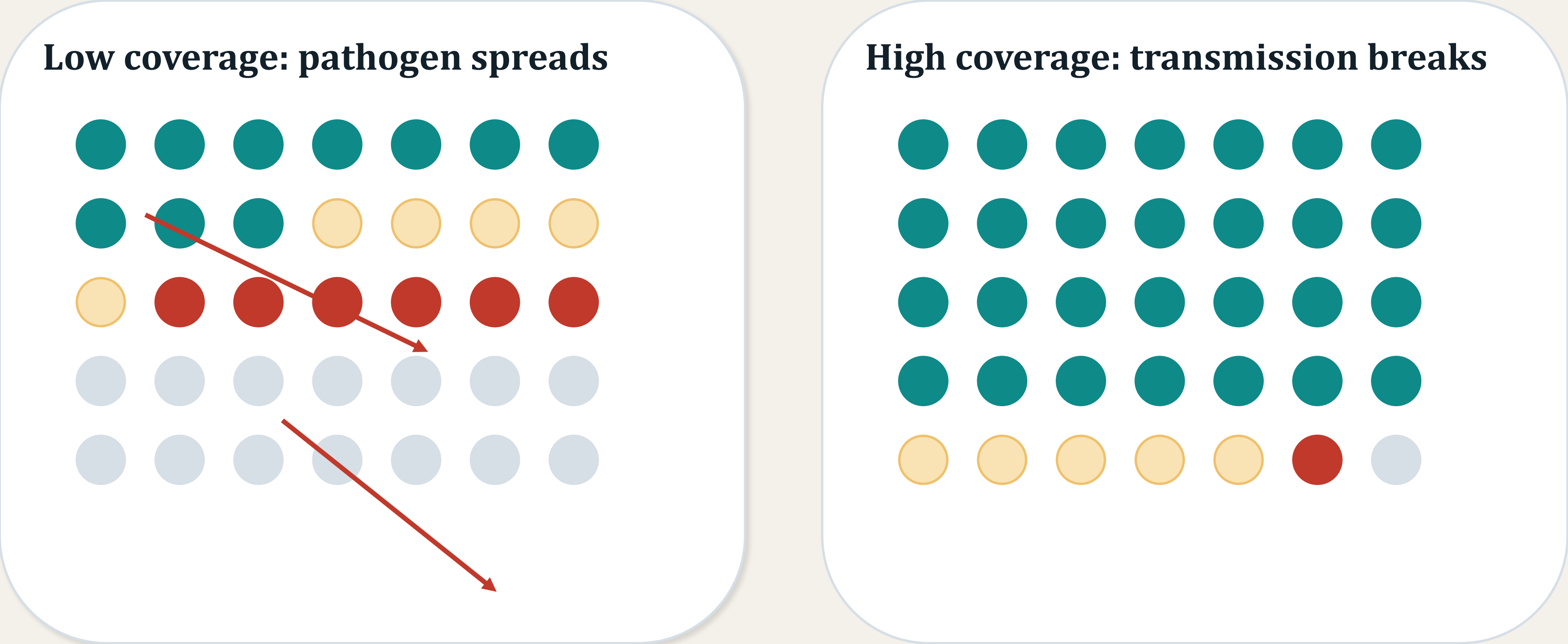
Why smallpox was suited for eradication: human-only virus • distinctive symptoms • cheap, effective vaccine

Source: Our World in Data (ourworldindata.org/vaccination)

Figure reference: US vaccine-preventable disease reductions (Roush & Murphy, 2007)

Herd Immunity: How Vaccinating Many Protects the Most Vulnerable

Herd immunity helps shield newborns, immunocompromised individuals, and cancer patients on chemotherapy who cannot be vaccinated.



Vaccinated

Cannot be vaccinated

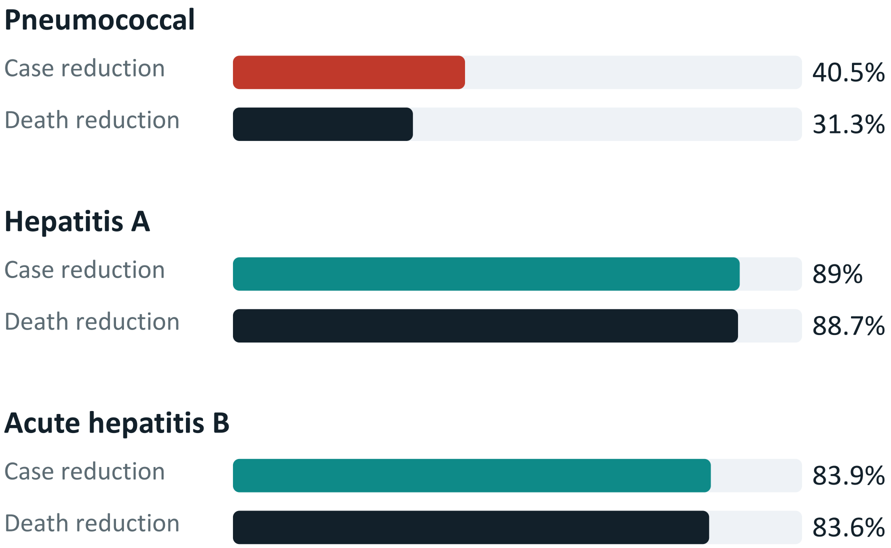
Infected / infectious

Indirectly protected

Source: Our World in Data (ourworldindata.org/vaccination)

Gaps Persist: Pneumococcal Disease Shows Only 40% Case Reduction

Incomplete disease control (US reductions)



Why gaps persist (from brief)

- Not everyone has access to vaccines
- Supply shortages put children at risk
- Poor scientific understanding can reduce trust and uptake

Policy focus: sustain high coverage + close access gaps to protect those who cannot be vaccinated.

Source: Our World in Data (ourworldindata.org/vaccination)

Key Takeaways — and Priorities

Elimination is achievable

100% case + death reduction in the US for diphtheria, smallpox, and acute polio.

Development is accelerating

From 105 years (typhoid) to <1 year (COVID-19) from pathogen identification to licensure.

Coverage changes outcomes

Measles fell to near zero across all states after the 1980 kindergarten mandate.

Gaps still matter

Pneumococcal disease shows only 40.5% case reduction; hepatitis A and B remain incomplete.

Call to action for health systems

- Maintain high coverage to sustain elimination
- Close access and supply gaps globally
- Invest in trust, communication, and uptake

Source: Our World in Data (ourworldindata.org/vaccination)